

EQUIPMENT INSPECTION REPORT: CENTRIFUGAL PUMP P-101

ConfigIQ Sovereign AI Workbench — Equipment Integrity Inspection

Parameter	Measured Value	Design / SOP Threshold	Condition Status
Vibration RMS (Outboard)	7.4 mm/s	<= 4.5 mm/s (Trip > 7.1)	CRITICAL EXCEEDED
Bearing Temperature	88.0 deg C	<= 75.0 deg C (SOP-M-104)	EXCEEDED HIGH
Mechanical Seal Leakage	14 drops/min	<= 5 drops/min	EXCEEDED LEAK
Suction Pressure	0.32 MPa	0.30 - 0.40 MPa	NORMAL
Discharge Pressure	1.85 MPa	1.80 - 2.10 MPa	NORMAL
Flow Rate (Q)	48.5 m3/h	50.0 m3/h BEP	SLIGHT REDUCTION

Key Technical Findings:

- Outboard bearing vibration reached 7.4 mm/s RMS, exceeding ISO 10816-3 Zone D trip limit of 7.1 mm/s.
- Bearing running temperature elevated to 88.0 deg C, exceeding SOP-M-104 ceiling of 75.0 deg C.
- Silicon carbide mechanical cartridge seal dynamic leakage measured at 14 drops/minute (> 5 drops/min limit).
- Localized pitting and cavitation erosion detected on impeller suction eye (depth: 1.2 mm).

Risk Severity Assessment: *CRITICAL — IMMEDIATE ACTION REQUIRED*

Inspector Corrective Recommendations:

- Immediately execute emergency shutdown and electrical LOTO isolation per SOP-SAF-001.
- Dismantle pump casing, replace mechanical cartridge seal assembly.
- Replace drive-end and non-drive-end bearing assemblies.
- Perform precision laser shaft alignment (angular offset <= 0.05 mm, radial <= 0.05 mm).
- Conduct hydrostatic pressure validation test at 1.5x MAWP for 15 minutes before recommissioning.